



BUBT | BANGLADESH UNIVERSITY OF
BUSINESS AND TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

LAB REPORT
IF-ELSE PROBLEMS

Microprocessor and Microcontroller Lab (CSE-326)

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June 16, 2026

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1 Alphabetic order

1.0.1 Question

1. Take input two letters and print them in alphabetical order.

Sample input/output screen:

```
Enter the first letter: H
Enter the second letter: A
The letters in alphabetical order: AH
```

1.0.2 Solution

```
1 .MODEL SMALL
2 .STACK 100H
3
4 .DATA
5 STR DB "Enter the first letter: $"
6 STR1 DB 10,13,"Enter the second letter: $"
7 STR2 DB 10,13,"The letters in alphabetical order: $"
8
9 .CODE
10 MAIN PROC
11     MOV AX, @DATA
12     MOV DS, AX
13
14     MOV AH, 9
15     LEA DX, STR
16     INT 21H
17
18     MOV AH, 1
19     INT 21H
20     MOV BL, AL
21
22     MOV AH, 9
23     LEA DX, STR1
24     INT 21H
25
26     MOV AH, 1
27     INT 21H
28     MOV CL, AL
29
30     MOV AH, 9
31     LEA DX, STR2
32     INT 21H
33
34     CMP BL, CL
35     JLE Order
```

```

36
37     MOV AH, 2
38     MOV DL, CL
39     INT 21H
40
41     MOV DL, BL
42     INT 21H
43     JMP Exit
44
45 Order:
46     MOV AH, 2
47     MOV DL, BL
48     INT 21H
49
50     MOV DL, CL
51     INT 21H
52
53 Exit:
54     MOV AH, 4CH
55     INT 21H
56
57 MAIN ENDP
58 END MAIN

```

Listing 1: Take input two letters and print them in alphabetical order.

Output

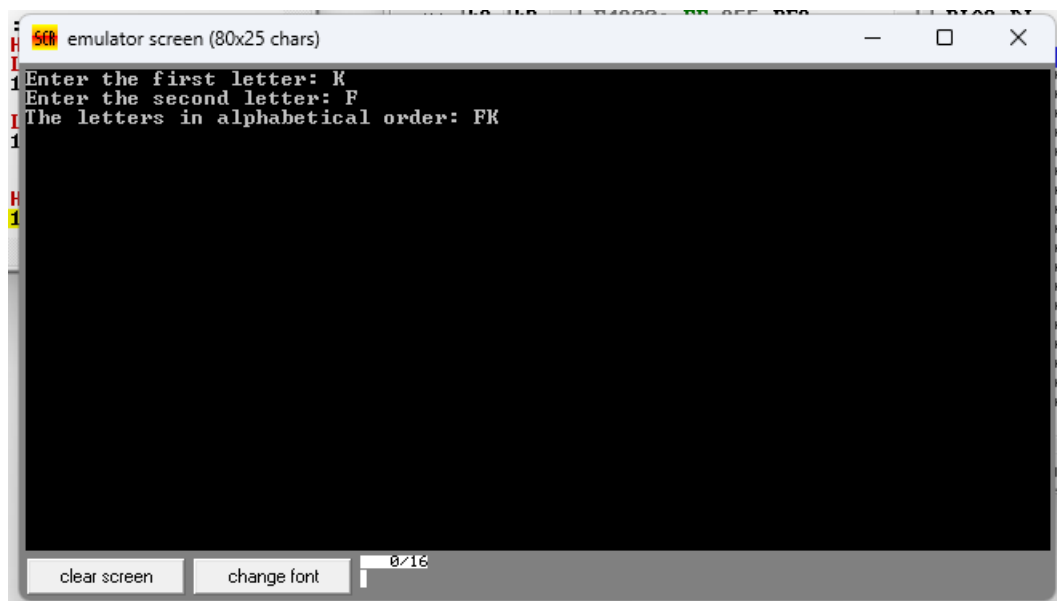


Figure 1: Task 1

“latex

2 Reverse alphabetic order

2.0.1 Question

1. Take input two letters and print them in reverse alphabetical order.

Sample input/output screen:

```
Enter the first letter: H
Enter the second letter: A
The letters in reverse alphabetical order: HA
```

2.0.2 Solution

```
1  .MODEL SMALL
2  .STACK 100H
3
4  .DATA
5  STR  DB "Enter the first letter: $"
6  STR1 DB 10,13,"Enter the second letter: $"
7  STR2 DB 10,13,"The letters in reverse alphabetical order: $"
8
9  .CODE
10 MAIN PROC
11     MOV AX, @DATA
12     MOV DS, AX
13
14     MOV AH, 9
15     LEA DX, STR
16     INT 21H
17
18     MOV AH, 1
19     INT 21H
20     MOV BL, AL
21
22     MOV AH, 9
23     LEA DX, STR1
24     INT 21H
25
26     MOV AH, 1
27     INT 21H
28     MOV CL, AL
29
30     MOV AH, 9
31     LEA DX, STR2
32     INT 21H
33
34     CMP BL, CL
35     JGE Order
```

```

36
37     MOV AH, 2
38     MOV DL, CL
39     INT 21H
40
41     MOV DL, BL
42     INT 21H
43     JMP Exit
44
45 Order:
46     MOV AH, 2
47     MOV DL, BL
48     INT 21H
49
50     MOV DL, CL
51     INT 21H
52
53 Exit:
54     MOV AH, 4CH
55     INT 21H
56
57 MAIN ENDP
58 END MAIN

```

Listing 2: Take input two letters and print them in reverse alphabetical order.

Output

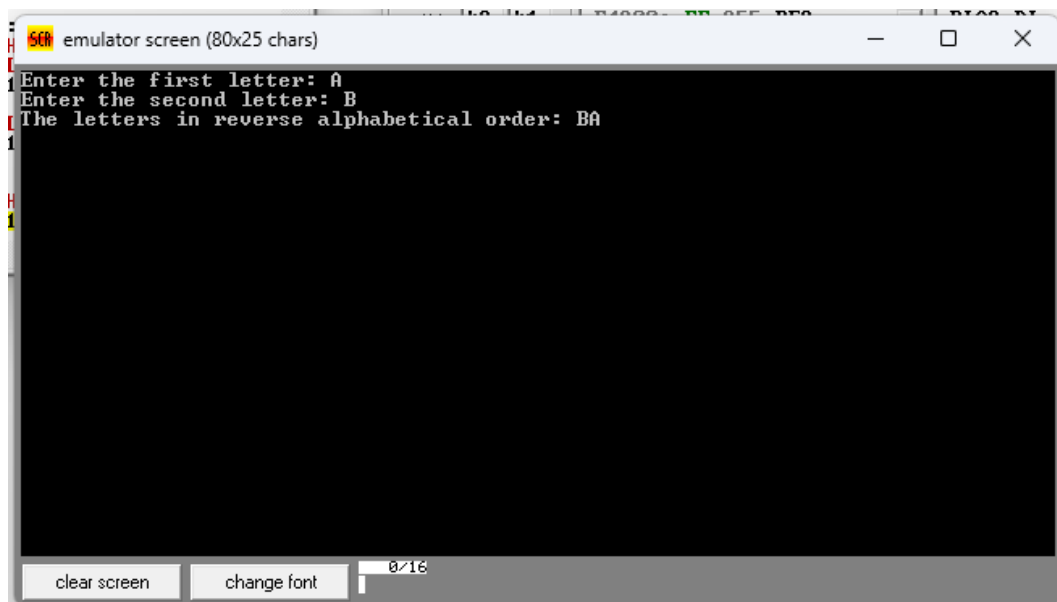


Figure 2: Task 2

3 Vowel or consonant

3.0.1 Question

1. Take letter input and check if it is a vowel or not.

Sample input/output screen:

```
Enter letter: A
It is a vowel.
```

```
(another example)
Enter letter: K
It is a consonant.
```

3.0.2 Solution

```
1 .MODEL SMALL
2 .STACK 100H
3
4 .DATA
5 STR DB "Enter letter: $"
6 STR1 DB 10,13,"It is a vowel$"
7 STR2 DB 10,13,"It is a consonant$"
8
9 .CODE
10 MAIN PROC
11     MOV AX, @DATA
12     MOV DS, AX
13
14     MOV AH, 9
15     LEA DX, STR
16     INT 21H
17
18     MOV AH, 1
19     INT 21H
20     MOV BL, AL
21
22     CMP BL, 'A'
23     JE Vowel
24     CMP BL, 'E'
25     JE Vowel
26     CMP BL, 'I'
27     JE Vowel
28     CMP BL, 'O'
29     JE Vowel
30     CMP BL, 'U'
31     JE Vowel
32
```

```

33     CMP BL, 'a'
34     JE Vowel
35     CMP BL, 'e'
36     JE Vowel
37     CMP BL, 'i'
38     JE Vowel
39     CMP BL, 'o'
40     JE Vowel
41     CMP BL, 'u'
42     JE Vowel
43
44     MOV AH, 9
45     LEA DX, STR2
46     INT 21H
47     JMP EXIT
48
49 Vowel:
50     MOV AH, 9
51     LEA DX, STR1
52     INT 21H
53
54 EXIT:
55     MOV AH, 4CH
56     INT 21H
57
58 MAIN ENDP
59 END MAIN

```

Listing 3: Take letter input and check if it is a vowel or not.

Output

(next page)



Figure 3: Task 3

4 Alphabet or digit

4.0.1 Question

1. Take a character input and check whether it is alphabet or digit.

Sample input/output screen:

```
Enter character: A
Alphabet
```

```
(another example)
Enter character: 7
Digit
```

4.0.2 Solution

```

1  .MODEL SMALL
2  .STACK 100H
3
4  .DATA
5  STR  DB "Enter a character: $"
6  STR1 DB 10,13,"Digit$"
7  STR2 DB 10,13,"Alphabet$"
8  STR3 DB 10,13,"Invalid character$"
9
10 .CODE
11 MAIN PROC
12     MOV AX, @DATA
13     MOV DS, AX
14
15     MOV AH, 9
16     LEA DX, STR
17     INT 21H
18
19     MOV AH, 1
20     INT 21H
21     MOV BL, AL
22
23     ; check digit 0-9
24     CMP BL, '0'
25     JL CHECK_UPPER
26     CMP BL, '9'
27     JLE DIGIT
28
29 CHECK_UPPER:
30     ; check uppercase A-Z
31     CMP BL, 'A'
32     JL INVALID
33     CMP BL, 'Z'
34     JLE ALPHABET
35
36 CHECK_LOWER:
37     ; check lowercase a-z
38     CMP BL, 'a'
39     JL INVALID
40     CMP BL, 'z'
41     JLE ALPHABET
42
43     JMP INVALID
44
45 DIGIT:
46     MOV AH, 9
47     LEA DX, STR1
48     INT 21H
49     JMP EXIT
50

```

```
51 ALPHABET:
52     MOV AH, 9
53     LEA DX, STR2
54     INT 21H
55     JMP EXIT
56
57 INVALID:
58     MOV AH, 9
59     LEA DX, STR3
60     INT 21H
61
62 EXIT:
63     MOV AH, 4CH
64     INT 21H
65
66 MAIN ENDP
67 END MAIN
```

Listing 4: Take a character input and check whether it is alphabet or digit.

Output



Figure 4: Task 4

5 Smallest among three digits

5.0.1 Question

1. Find out the smallest among 3 digits.

Sample input/output screen:

```
Enter first value: 8
Enter second value: 1
Enter third value: 6
The smallest value is: 1
```

5.0.2 Solution

```
1 .MODEL SMALL
2 .STACK 100H
3
4 .DATA
5 STR DB "Enter first digit: $"
6 STR1 DB 10,13,"Enter second digit: $"
7 STR2 DB 10,13,"Enter third digit: $"
8 STR3 DB 10,13,"Smallest: $"
9 STR4 DB 10,13,"Largest: $"
10
11 .CODE
12 MAIN PROC
13     MOV AX, @DATA
14     MOV DS, AX
15
16     MOV AH, 9
17     LEA DX, STR
18     INT 21H
19
20     MOV AH, 1
21     INT 21H
22     MOV BL, AL
23
24     MOV AH, 9
25     LEA DX, STR1
26     INT 21H
27
28     MOV AH, 1
29     INT 21H
30     MOV CL, AL
31
32     MOV AH, 9
33     LEA DX, STR2
34     INT 21H
```

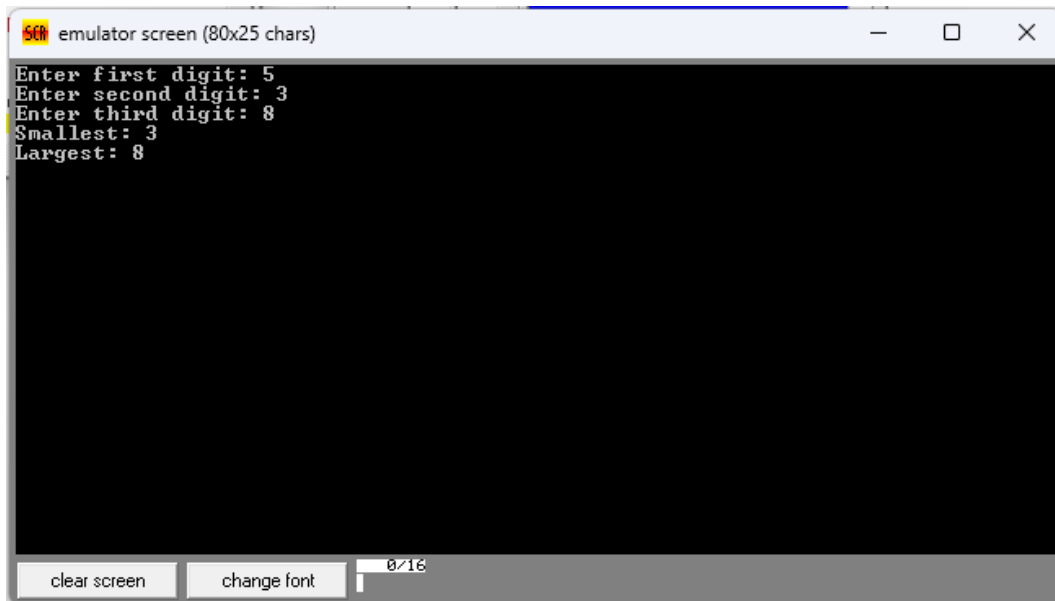
```

35
36     MOV AH, 1
37     INT 21H
38     MOV CH, AL
39
40     ; assume BL = smallest, CL = middle, CH = largest
41
42     CMP BL, CL
43     JLE STEP1
44     MOV DL, BL
45     MOV BL, CL
46     MOV CL, DL
47
48 STEP1:
49     CMP CL, CH
50     JLE STEP2
51     MOV DL, CL
52     MOV CL, CH
53     MOV CH, DL
54
55 STEP2:
56     CMP BL, CL
57     JLE PRINT
58     MOV DL, BL
59     MOV BL, CL
60     MOV CL, DL
61
62 PRINT:
63     MOV AH, 9
64     LEA DX, STR3
65     INT 21H
66
67     MOV AH, 2
68     MOV DL, BL
69     INT 21H
70
71     MOV AH, 9
72     LEA DX, STR4
73     INT 21H
74
75     MOV AH, 2
76     MOV DL, CH
77     INT 21H
78
79 EXIT:
80     MOV AH, 4CH
81     INT 21H
82
83 MAIN ENDP
84 END MAIN

```

Listing 5: Find out the smallest among 3 digits.

Output



```
emulator screen (80x25 chars)
Enter first digit: 5
Enter second digit: 3
Enter third digit: 8
Smallest: 3
Largest: 8
```

The screenshot shows a window titled "emulator screen (80x25 chars)". The text inside the window is as follows:

```
Enter first digit: 5
Enter second digit: 3
Enter third digit: 8
Smallest: 3
Largest: 8
```

At the bottom of the window, there are two buttons labeled "clear screen" and "change font", and a small status bar showing "0/16".

Figure 5: Task 5

1 2

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